

Incoming SCADA Telemetry Stream

Inputs: Wind Speed v , Active Power P , Blade Pitch β , Timestamp t

Regimes: Clean ($\tau = 0$) | Degraded ($\tau \leq 20$ m) | Contingency ($\beta = \emptyset$, $\tau = 60$ m)



Observability & Staleness Gate

Audits Pitch Register $\exists\beta$ and Timestamp Freshness τ

$\tau = 0, \exists\beta$

$\tau \leq 20$ m, $\exists\beta$

$\beta = \emptyset$ or $\tau \geq 60$ m

Tier 1: Physical

Continuous Quantile

- Nominal $\tau = 0$
- 6.8% tail violation
- Zero ML overhead
- IEC 61400 rule

[Physics Optimal]

Tier 2: Recalibrate

State-Conditional

- $\tau \in [10, 20]$ min
- Pitch retained ($\exists\beta$)
- Absorbs 55.3% loss
- Empirical $q_\tau(v)$

[Recalibration]

Tier 3: STGQ-Mod

Residual Head

- $\beta = \emptyset$ or $\tau \geq 60$ m
- 9.7% tail violation
- Saves 46k–68k kWh
- Latent state recovery

[Representation]